

# WNC Salamanders - Practice Data Notes

The SQL file is a small teaching dataset, not an ecological survey or a complete species inventory. The species and broad habitat choices are based on the North Carolina Wildlife Resources Commission pages below. Descriptions are newly written summaries. No wildlife photographs or copied species-profile text are included.

| Common name                     | Scientific name used in the SQL                   | Source                                                       |
|---------------------------------|---------------------------------------------------|--------------------------------------------------------------|
| Eastern Newt                    | <i>Notophthalmus viridescens</i>                  | <a href="#">NC Wildlife: Eastern Newt</a>                    |
| Spotted Salamander              | <i>Ambystoma maculatum</i>                        | <a href="#">NC Wildlife: Spotted Salamander</a>              |
| Blue Ridge Two-lined Salamander | <i>Eurycea wilderae</i>                           | <a href="#">NC Wildlife: Blue Ridge Two-lined Salamander</a> |
| Red Salamander                  | <i>Pseudotriton ruber</i>                         | <a href="#">NC Wildlife: Red Salamander</a>                  |
| Eastern Hellbender              | <i>Cryptobranchus alleganiensis alleganiensis</i> | <a href="#">NC Wildlife: Eastern Hellbender</a>              |

For the Eastern Hellbender, the source abbreviates the name as *Cryptobranchus a. alleganiensis*; the SQL expands the subspecies name. The summary calls these five species records for classroom convenience; this entry names the eastern subspecies.

## Broad habitat categories

*Woodlands*, *Streams and rivers*, and *Ponds and pools* are categories chosen for this exercise. Links can represent use at different life stages or seasons, not continuous occupation of every habitat by every individual. For example, a newt's terrestrial eft and aquatic adult stages make it useful for a many-to-many example.

The seed relationships are:

| Record                          | Habitat links                 |
|---------------------------------|-------------------------------|
| Eastern Newt                    | Woodlands; Ponds and pools    |
| Spotted Salamander              | Woodlands; Ponds and pools    |
| Blue Ridge Two-lined Salamander | Woodlands; Streams and rivers |
| Red Salamander                  | Woodlands; Streams and rivers |
| Eastern Hellbender              | Streams and rivers            |

That gives **5 salamander records**, **3 habitat records**, and **9 links**. This is our illustrative data model; it is not a claim that these are the only habitats a species can use. The habitat descriptions are deliberately broad.

The database is separate from `wnc_birds`. Both projects can remain in use. Additional animal data, model classes, and pages can be added in later lessons without replacing these three in-class labs.

## Technical references

The Markdown labs link directly to the relevant PHP manual sections. SQL uses InnoDB foreign keys with parent-to-link cascades; the [MySQL foreign-key reference](#) explains those actions. No images, file uploads, or filesystem deletion are involved here.